Catalyst.jl

What is Catalyst.jl?

A Julia package for building and simulating reaction-network models.

- Describe how quantities change through interactions and rates
- Originally focused on chemical reactions
- Also useful for ecology, epidemiology, and systems biology

How does it relate to numerical computation?

- ODE solvers: simulate how a system changes over time
- Rootfinding: find steady states / equilibria
- Jacobians and linear algebra: support solver calculations
- Optimization: estimate model parameters from data

Catalyst specifies the model; SciML solvers do the numerical work.

Who uses it, and what is in it?

Users

- Computational biologists
- Chemists and systems biologists
- Researchers modeling populations or epidemics

Software

- Catalyst: reaction network definitions
- ModelingToolkit / Symbolics: symbolic models
- OrdinaryDiffEq / JumpProcesses: simulations

Demo: simulate disease spread

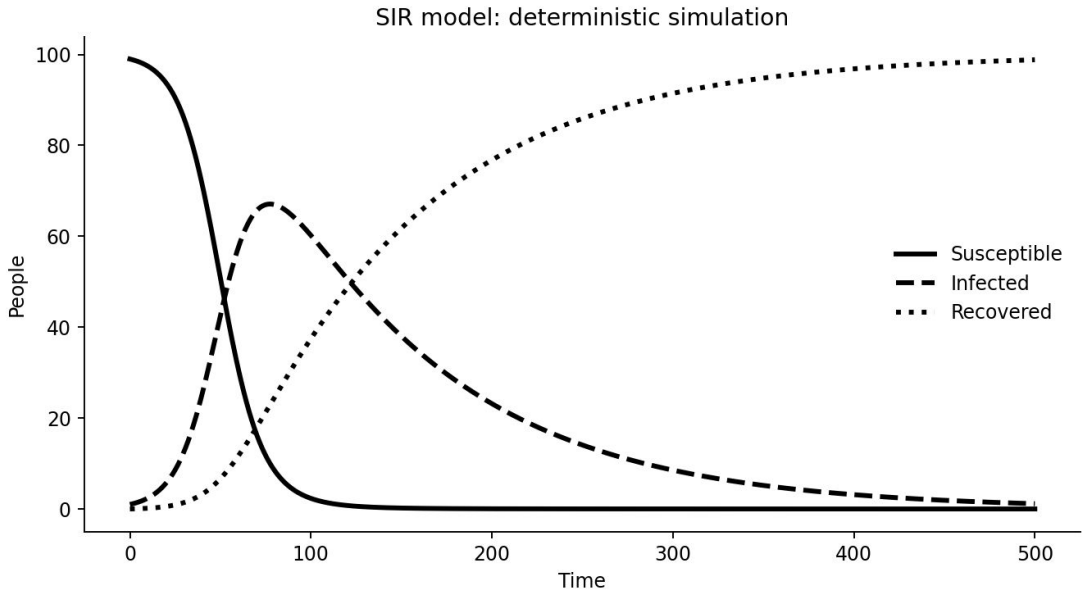
$S + I \rightarrow 2I \qquad I \rightarrow R$

Define a reaction network in Julia:

```
using Catalyst, OrdinaryDiffEq

sir = @reaction_network begin
    alpha, S + I --> 2I
    beta, I --> R
end

u0 = [:S => 99, :I => 1, :R => 0]
p = [:alpha => 0.001, :beta => 0.01]
prob = ODEProblem(sir, u0, (0., 500.), p)
sol = solve(prob)
```



Illustrative plot of the same ODE model

Possible group projects

- Write a tutorial or reproducible demo for Catalyst v16
- Create worked examples for upgrading from v15 to v16
- Resolve a small issue or add correctness tests
- Benchmark a simple model using different numerical methods

Goal: a small, useful contribution we can submit upstream.